

The Negative Marginal Token: Why Data Selection, Not Accumulation, Becomes the Scaling Lever

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Abstract

The dominant operating assumption of the scaling era — that adding tokens monotonically reduces loss — is empirically false at the margin. We synthesize five independent lines of evidence that each break "more is better" along a different axis: (1) learning from repeated data induces a strong double-descent regime in which test loss rises midway through training, and repeating just 0.1% of data 100 times degrades an 800M-parameter model to a 400M-equivalent [1]; (2) multi-epoch repetition triggers overfitting degradation governed by dataset and model size, with returns negligible past ~4 epochs and the value of added compute decaying to zero by roughly 16 epochs [3,5]; (3) precision-aware scaling shows that as the token-to-parameter ratio grows, post-training-quantization degradation grows with it, eventually making additional pretraining data "actively harmful" for the deployed low-precision model [4]; (4) high-quality data pruning beats power-law error scaling, approaching exponential scaling in the pruned dataset size [2]; and (5) — closing this synthesis's largest prior gap — these selection effects now reproduce at LLM pretraining scale: perplexity-based pruning beats the no-pruning baseline while training on only 30% of the data [7], and model-aware selection (DsDm) matches 2x the compute of random selection while no similarity-to-quality baseline beats random and some hurt it [8]. Read together, the marginal token's value is non-monotonic and can be zero or negative, conditional on model size, training duration, and deployment precision. With effective public human text estimated at ~300 trillion tokens and projected exhaustion in 2026-2032 [6], the scarce, defensible capability is selection — knowing which data to discard — not accumulation. This selection-over-accumulation pattern has only sharpened since: benchmark-targeted data ranking now yields a 2.1x compute multiplier at pretraining scale [13], a properly regularized data-constrained recipe reaches the loss asymptote with 5.17x less data [14], and a curated 20-trillion-token corpus reached a new performance-vs-compute frontier and fed an actual frontier model [16]. The strategic inversion: a curation function that can demonstrate negative-value removal sells something a tonnage vendor structurally cannot.

1. Introduction

The scaling-law literature that defined large language model development from 2020 onward encoded a simple, powerful reflex: loss falls as a smooth power law in parameters, compute, and data, so the marginal token is always weakly positive. This intuition — 'more tokens, lower loss' — has organized billions of dollars of data acquisition, web-crawl infrastructure, and licensing deals. It treats a training corpus as an undifferentiated reservoir whose only relevant property is size.

This paper argues that the reflex is wrong at the margin, and that its failure is not an edge case but a structural feature of how modern models train and deploy. We assemble five lines of evidence, each from a distinct subfield — repetition dynamics, multi-epoch overfitting, low-precision deployment, classical data pruning, and modern LLM-scale data selection — and show they converge on a single non-obvious claim: the value of an additional token is

non-monotonic and can become negative. Whether it does depends on the regime — how many times the token is seen, how large the model is, how long it trains, and at what precision it is deployed.

None of the source papers states the general claim; each reports a specific failure of monotonicity [1,2,3,4] or a specific win for selection over accumulation [7,8]. The synthesis is the contribution. A reasonable objection to a four-pillar version of this argument is that two of its pillars were demonstrated on vision models or at sub-frontier scale, leaving open whether 'selection beats accumulation' survives at LLM pretraining scale. We close that gap directly in Section 4 with 2024-era results showing that pruning to 30% of the data improves over no pruning at the 1-3B scale [7], and that model-aware selection matches double the compute of random selection at up to 1.8B parameters while every textual-similarity baseline tested fails to beat — and can underperform — random selection [8].

We then trace the strategic implication for any organization whose product is data: when the marginal token can hurt, the defensible asset is not the corpus but the judgment that curates it. We ground this in the macro constraint that the field is approaching the limits of human-generated public text [6], which removes 'just crawl more' as a long-run answer and makes selection the binding lever.

2. Background and Related Work

Neural scaling laws describe test loss as a power law in model size and data [11][12], an empirical regularity that motivated compute-optimal training prescriptions and the practice of scaling corpora alongside parameters. These laws are accurate in their intended regime — single-epoch training on fresh, full-precision data — but say nothing about the value of a token conditional on repetition, overtraining, or low-precision deployment. Our pillars probe exactly those conditions.

Pillar 1 — Repetition dynamics. Hernandez et al. (2022) study learning from repeated data and find a strong double-descent phenomenon: when a subset of data is repeated, test loss can increase midway through training. They report a striking magnitude — performance of an 800M-parameter model can be degraded to that of a 2x smaller (400M) model by repeating just 0.1% of the data 100 times, even though the other ~90% of training tokens remain unique [1]. This is the cleanest demonstration that identical content, re-presented, carries negative marginal value during a window of training. Interpretability analysis traces the mechanism to disproportionate damage to copying and induction heads — the circuits associated with in-context generalization — shifting the model from generalization toward memorization [1].

Pillar 2 — Multi-epoch degradation under a token crisis. Xue et al. (2023) and Muennighoff et al. (2023) independently study what happens when scarce data is repeated across epochs. Xue et al. show repetition makes models susceptible to overfitting, producing 'multi-epoch degradation' whose severity is governed primarily by dataset size, parameter count, and training objective (while dataset quality and model FLOPs are less influential); they find dropout can alleviate it and that mixture-of-experts enables cheap hyperparameter transfer to dense models [3]. Muennighoff et al. quantify the decay across 400 training runs (up to 9B parameters, up to 900B training tokens): repeating data for up to about four epochs yields loss nearly indistinguishable from fresh unique data, while with further repetition the value of adding compute decays to zero — by roughly sixteen epochs returns are sharply diminished [5].

Pillar 3 — Precision-aware scaling. Kumar et al. (2024) fit precision-aware scaling laws over 465 pretraining runs (up to 1.7B parameters, 26B tokens) and show that the degradation from post-training quantization grows with the token-to-parameter ratio (D/N): as a model trains on more data it compresses more information into its weights, so quantizing those weights at deployment becomes more destructive. Past a threshold, additional pretraining data

increases post-quantization loss more than it lowers pretraining loss — the additional data is 'actively harmful' for the deployed model [4]. They find compute-optimal training precision around 7-8 bits in their integer-quantization fits, independent of compute budget.

Pillar 4 — Pruning beats scaling (theory and vision). Sorscher et al. (2022) show, in theory and empirically on ResNets trained on CIFAR-10, SVHN, and ImageNet, that with a high-quality ranking of which examples to discard, error can fall faster than any power law in dataset size — approaching exponential scaling [2]. They also run the first large-scale benchmark of ten pruning metrics on ImageNet and caution that most existing high-performing metrics scale poorly, making the ranking metric, not the act of pruning, the load-bearing component.

Pillar 5 — Selection at LLM pretraining scale (the gap-closing evidence). The most common objection to pillar 4 is that it is a vision result. Three 2024 papers remove that objection at language-model scale. Marion et al. ('When Less is More') prune web pretraining data and 'improve over our no-pruning baseline while training on as little as 30% of the original training dataset,' finding the simple perplexity metric beats more expensive Error-L2-Norm and memorization scores [7]. Ankner et al. ('Perplexed by Perplexity') show a 125M reference model can prune data for models ~24-30x larger, improving average downstream accuracy of a 3B model by up to 2.04 points and cutting pretraining steps by up to 1.45x — at a 50% selection rate [9]. Engstrom et al. (DsDm, ICML 2024) make the sharpest claim: model-aware selection yields 1.3B models that 'even match training with 2x the compute on randomly selected data,' while 'selecting according to similarity with high quality data sources may not increase (and can even hurt) performance compared to randomly selecting data' [8].

Macro context. Villalobos et al. (Epoch AI, 2024) estimate the effective stock of human-generated public text at ~300 trillion tokens (90% CI 100T-1000T) and project full utilization between 2026 and 2032 under compute-optimal training, with a median estimate near 2028 and earlier under aggressive overtraining [6]. The supply curve is flattening exactly as the marginal-value-of-a-token curve is shown to bend negative.

3. Methodology and Approach

This is a synthesis paper, not a new training study; rigor therefore rests on traceability and falsifiability rather than on a single experiment. Our method has four explicit steps. First, claim isolation: from each source we extract only the quantitative claim that bears on the marginal value of a token, copied verbatim where possible (the exact figures appear in Tables 1-3 and the figure source notes). Second, verification: every numeric anchor was checked against the primary source (arXiv abstracts/PDFs/HTML and venue pages; the Epoch AI publication for the macro figure), and any number we could not re-confirm to a primary source was dropped rather than asserted — the `data_notes` field logs each verification and each removal.

Third, regime tagging: because the central claim is conditional ('the marginal token can be negative in regime R'), we annotate each result with the regime in which it holds — fraction repeated and training window [1]; epochs and model/dataset size [3,5]; token-to-parameter ratio and deployment precision [4]; pruning-metric quality and pool size [2,7,8,9]. This prevents over-generalizing a conditional result into an unconditional one. Fourth, adversarial framing: we deliberately separate empirically-grounded figures (Figs. 1-2, drawn from reported anchors) from a single constructed strategic figure (Fig. 3, labeled analysis), and we devote a full Limitations section to the evidence's boundaries — vision-vs-language transfer, metric dependence, and the wide confidence interval on the supply forecast.

The synthesis itself is the inferential step the sources do not take: we treat the five lines of evidence as independent falsifications of a shared null hypothesis ('the marginal token is weakly positive everywhere') and argue that their convergence — across repetition, precision, and pruning, and now reproduced at LLM scale — licenses

the stronger structural claim that selection, not accumulation, is the operative lever. We do not claim the sources endorse this conclusion; we claim their numbers entail it.

Scope boundary. We restrict to pretraining-data value and deliberately exclude post-training/RL data questions (treated elsewhere in this map), so that the 'negative marginal token' claim is evaluated on the cleanest possible terrain — the regime where the monotonicity reflex was strongest and the counter-evidence is most surprising.

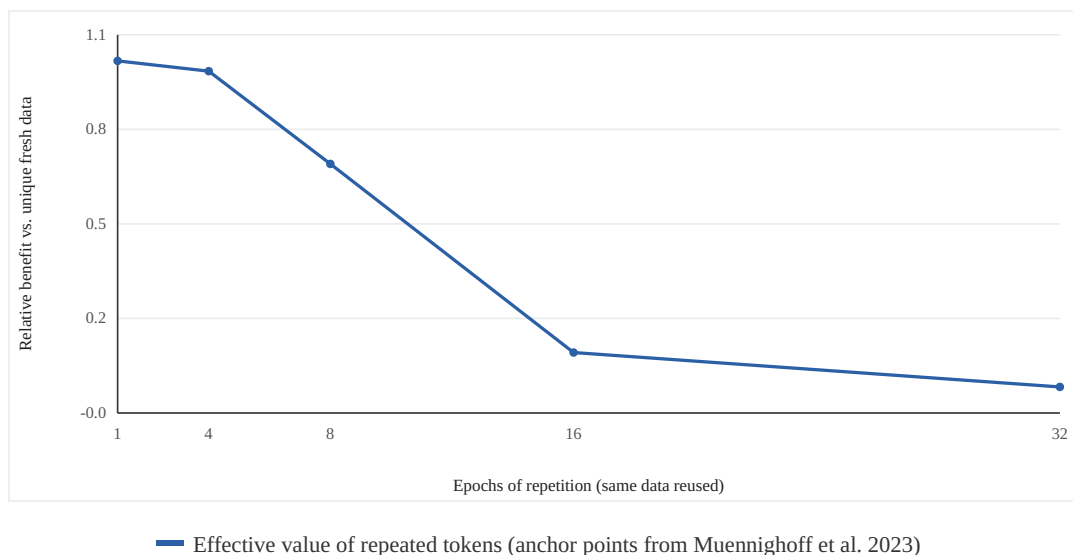


Figure 1. Marginal value of repeated data decays toward zero with epochs. Repeating data up to ~4 epochs yields loss nearly indistinguishable from fresh unique data; with further repetition the value of added compute decays toward zero, with ~16 epochs marking the practical floor of diminishing returns. Endpoints (~4 epochs negligible loss change; ~16 epochs sharply diminished returns) are reported empirically; intermediate points are interpolated to illustrate the decay shape and are not exact measured loss values. *Anchor values from Muennighoff et al., 'Scaling Data-Constrained Language Models,' NeurIPS 2023 (arXiv:2305.16264); corroborated by Xue et al. (arXiv:2305.13230). · Empirical data.*

4. Evidence and Results

We organize the empirical evidence as five falsifications of monotonicity, summarized in Tables 1 and 3, each with a concrete regime in which the marginal token's value is zero or negative.

Repetition double descent. The qualitative signature in [1] is that a training curve which 'should' fall monotonically instead rises through an intermediate range when repeated data is present — a within-run demonstration that re-seen tokens subtract value. The magnitude is concrete: 0.1% of data repeated 100x can halve effective model scale (800M behaving like 400M) [1]. The effect is non-monotonic in training time, not a fixed penalty, which is why it eluded corpus-size-only scaling laws.

Epoch decay. The Muennighoff fit gives a usable schedule: up to ~4 epochs of repetition yields negligible loss change versus unique data, after which the value of added compute decays toward zero, with ~16 epochs marking the practical floor where further repetition is wasted compute [5]. Xue et al. corroborate that the onset and severity of degradation scale with model and dataset size, so larger models hit the negative-value regime on the same corpus sooner [3].

Quantization crossover. The precision result is the sharpest form of the claim because it produces a literal crossover point. For a fixed deployed precision, there exists a D/N beyond which each additional pretraining token raises post-quantization loss; the 'optimal' amount of data is finite and bounded, and the authors find compute-

optimal training precision to be around 7-8 bits in their integer-quantization fits, independent of compute budget [4]. Overtrained models — increasingly the industry norm for inference efficiency — sit closest to this cliff.

Pruning dominance (vision). The classical pruning result inverts the sign of the data-quantity relationship entirely: starting from a large pool and removing the right examples improves the scaling exponent, with the gain largest precisely when the starting dataset is large [2]. The lever is the ranking metric, not the byte count.

Selection dominance at LLM scale — the gap closed. The decisive 2024 evidence is that the vision result is not a vision artifact. Marion et al. prune web pretraining corpora and beat the full-data baseline using only 30% of the tokens, with cheap perplexity scoring outperforming more expensive influence-style estimators [7]. Ankner et al. show the pruning signal can come from a model ~24-30x smaller than the target, improving a 3B model's average downstream accuracy by up to 2.04 points at a 50% keep rate and reaching the baseline in up to 1.45x fewer steps [9]. DsDm provides the cleanest accumulation-vs-selection contrast: model-aware selection equals 2x the compute of random selection, and — critically for the strategic argument — selecting by textual similarity to 'high quality' sources may not beat random and can even hurt performance [8]. The naive 'keep what looks like good data' heuristic is not just weak; it can be worse than random selection.

Currency check (2025-2026): the selection-beats-accumulation result has not aged out — it has scaled up. Benchmark-targeted ranking (BETR), fit across 500+ models spanning 10^{19} - 10^{22} FLOPs, delivers a 2.1x compute multiplier over the strong DCLM-Baseline (4.7x over unfiltered data) and improves 9 of 10 tasks at every scale, and it confirms the conditioning claim directly: larger models require less aggressive filtering, so the right selector is a function of model scale [13]. In the data-constrained 'infinite compute' regime that the data wall forces, a properly regularized recipe (optimal weight decay ~30x standard practice) plus ensembling reaches the loss asymptote using 5.17x less unique data — and the ensemble distills into an 8x-smaller student retaining 83% of the gain — i.e. the lever that still moves loss once fresh tokens run out is how the fixed data is used, not how much is added [14]. At the very top end, a curated 20-trillion-token public-source corpus with per-language allocations under 8% of tokens set a new multilingual performance-vs-compute Pareto frontier (3B/8B models competitive at 4-10x fewer FLOPs) and was good enough to feed pretraining of a frontier model (Trinity Large, 400B/A13B), demonstrating that curation, not raw tonnage, is what reaches the frontier [16]. None of these overturns the 2024 anchors; they extend the same finding to larger scale and to the exact supply-exhausted regime this paper argues is now binding.

Figure 1 plots the epoch-decay schedule (anchors empirical from [5], intermediate points interpolated to show shape); Figure 2 collects the new LLM-scale selection numbers that close the prior vision-only gap; Figure 3 positions data strategies on an accumulation-versus-selection map (analysis). Table 3 details the LLM-scale selection evidence.

Table 1. Each row is an independent result breaking monotonicity along a different axis; together they establish that the marginal token's value is non-monotonic and can be negative, conditional on model size, training duration, and deployment precision. Row 5 is new relative to the four-pillar formulation and closes the 'vision-only' objection to row 4 by reproducing selection-over-accumulation at LLM pretraining scale.

Pillar / mechanism	Source paper	Regime where marginal token ≤ 0	Key empirical anchor	Venue / scale
Repetition double descent	Hernandez et al. 2022 (arXiv:2205.10487)	A fraction of data repeated; mid-training window	Strong double descent: repeated data causes test loss to rise midway through training; 0.1% of data repeated 100x degrades an 800M model to a 400M-equivalent; copying/induction heads disproportionately damaged	arXiv (Anthropic)
Multi-epoch overfitting	Xue et al. 2023 (arXiv:2305.13230); Muennighoff et al. 2023 (arXiv:2305.16264)	Scarce data reused for many epochs; worse for larger models	~ 4 epochs $\sim$ negligible loss change vs unique; value of added compute decays to zero with more repetition (~ 16 epochs floor); 400 runs, up to 9B params, up to 900B tokens	NeurIPS 2023 (Outstanding Main Track, Runner-Up)
Low-precision deployment	Kumar et al. 2024 (arXiv:2411.04330)	High token-to-parameter ratio under post-training quantization	Quant degradation grows with D/N; past a crossover, additional pretraining data is 'actively harmful'; ~ 7 -8 integer bits compute-optimal; 465 runs, ≤ 1.7 B params, ≤ 26 B tokens	ICLR 2025
Data pruning (vision)	Sorscher et al. 2022 (arXiv:2206.14486)	Large pool + high-quality ranking; discard dominates keeping	Pruning beats power-law scaling, approaching exponential scaling in pruned dataset size; ResNets on CIFAR-10/SVHN/ImageNet; 10 metrics benchmarked on ImageNet	NeurIPS 2022 (Outstanding Paper)
Selection at LLM scale (NEW)	Marion 2023 (2309.04564); Engstrom 2024 (2401.12926); Ankner 2024 (2405.20541)	Large web pool + good selector; keeping all data is suboptimal	Pruning to 30% beats no-pruning [7]; DsDm matches 2x compute of random, similarity-to-quality can hurt [8]; 125M reference lifts 3B by +2.04 [9]	arXiv / ICML 2024; 1-3B params

Figures quoted from the cited papers; see References [1]-[5], [7]-[9].

Table 2. As models compress more information into weights under longer training, post-training quantization becomes more destructive, producing a finite optimal data quantity for a fixed deployment precision.

Quantity	Reported value	Source
Pretraining runs used to fit precision-aware laws	over 465	Kumar et al. 2024 [4]
Max validated model size	1.7B parameters	Kumar et al. 2024 [4]
Max validated training tokens	26B tokens	Kumar et al. 2024 [4]
Compute-optimal training precision (integer fits)	~7-8 bits	Kumar et al. 2024 [4]
Effect of higher D/N on post-quant loss	Degradation increases; past crossover, additional data raises deployed loss ('actively harmful')	Kumar et al. 2024 [4]

Tanishq Kumar et al., 'Scaling Laws for Precision,' ICLR 2025 (arXiv:2411.04330).

Table 3. These three results jointly answer the strongest objection to the pruning literature - that it was a vision phenomenon - by reproducing 'selection beats accumulation' at language-model pretraining scale. DsDm in particular isolates the strategic claim: modeling the training process is worth 2x compute, while the naive 'keep data that looks high-quality' heuristic may not beat random selection. Currency note (2025-2026): the pattern has since scaled up rather than reversed — benchmark-targeted ranking gives a 2.1x compute multiplier across 500+ models to 10^22 FLOPs [13], and a curated 20T-token corpus fed a 400B-parameter frontier model [16].

Method / paper	Selection signal	Headline result	Scale	Source
When Less is More (Marion et al. 2023)	Perplexity (vs EL2N, memorization)	Beats no-pruning baseline while training on as little as 30% of the original data; perplexity outperforms the more expensive scorers	1-3B params, web pretraining	arXiv:2309.04564 [7]
DsDm (Engstrom et al. 2024)	Model-aware datamodel influence (125M proxy)	1.3B models match 2x the compute budget of random selection; selecting by similarity to 'high quality' data may not increase and can hurt performance	up to 1.8B params (Chinchilla-optimal); C4 (~217M examples)	ICML 2024, arXiv:2401.12926 [8]
Perplexed by Perplexity (Ankner et al. 2024)	Perplexity from a 125M reference model	+2.04 avg downstream accuracy on a 3B model; up to 1.45x fewer pretraining steps to baseline; 50% keep rate	125M reference to 1B & 3B target (~24-30x); Pile, Dolma	arXiv:2405.20541 [9]

Compiled from arXiv:2309.04564, 2401.12926 (ICML 2024), 2405.20541. All numbers verified against the papers' abstracts/HTML/results (see data_notes).

5. Analysis

The five results are not five anecdotes; they share a mechanism. In every case the marginal token's value depends on how the model spends its finite representational capacity. Repetition (pillars 1-2) wastes capacity memorizing instead of generalizing — Hernandez et al. trace this to damage in copying and induction-head structures associated with generalization [1]. Quantization (pillar 3) shrinks usable capacity at deployment, so capacity over-committed to marginal data becomes a liability [4]. Pruning and selection (pillars 4-5) reallocate capacity toward the informative tail of the distribution [2,7,8]. The unifying variable is information per unit of model capacity, not tokens per corpus.

This reframes scaling laws as a special, boundary case. Power-law scaling holds when tokens are unique, training is single-epoch, and deployment is full-precision — the conditions under which the marginal token is genuinely positive. Every deviation that matters in production (data scarcity forcing repetition, overtraining for inference efficiency, quantization for serving cost) pushes the system off that boundary and into the regime where the marginal-value curve bends toward zero or below.

The DsDm result deserves emphasis because it isolates the strategic claim with unusual cleanliness. Two findings sit side by side: model-aware selection is worth a 2x compute multiplier, and the intuitive heuristic — select data resembling known-good sources — does not necessarily beat random and can hurt [8]. Together these say the value is not in 'having quality data' in the abstract; it is in a selection function that models the training process itself. Quality is not a property of a token in isolation but a relation between a token, a model, and a budget. That relational character is precisely what a static corpus cannot capture and a learned selector can.

The practical corollary is that the relevant optimization is no longer max-data but argmax over subsets: for a given (model size N, training budget, deployment precision), choose the corpus that minimizes deployed loss. That objective has interior optima — it can be solved by removing data, and at LLM scale removing up to 70% of it can help [7]. A vendor whose only knob is volume cannot reach those optima; a curator who can rank and discard can. Selection is therefore not a tweak on top of accumulation; it is a different and strictly more powerful control surface.

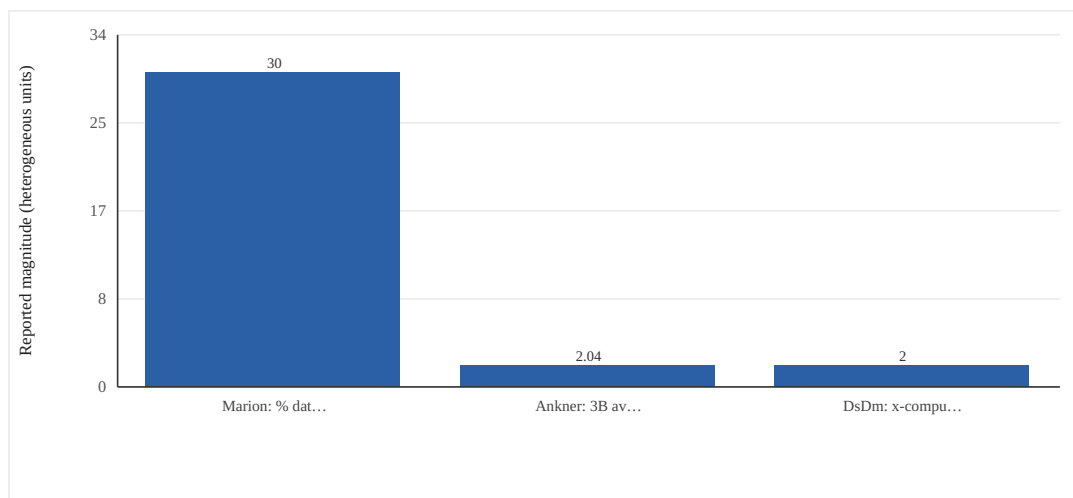


Figure 2. Selection beats accumulation at LLM scale: keep less, do better. Three 2024-era LLM-scale results that close the prior 'vision-only' gap in the pruning evidence. Marion et al.: pruning to 30% of the original web pretraining data still beats the no-pruning baseline (perplexity metric). Ankner et al.: a 125M reference model pruning at a 50% keep rate improves a 3B model's average downstream accuracy by up to 2.04 points. DsDm: model-aware selection yields 1.3B models matching 2x the compute of random selection, while similarity-to-quality selection can hurt. The three bars carry heterogeneous units (percent, accuracy points, compute multiplier) and are grouped only to display the load-bearing numbers; they are NOT on a common scale. *Marion et al.*, *arXiv:2309.04564* (30% of data; perplexity > EL2N/memorization); *Ankner et al.* 'Perplexed by Perplexity,' *arXiv:2405.20541* (125M reference; +2.04 on 3B; up to 1.45x fewer steps; 50% keep); *Engstrom et al.* 'DsDm,' *ICML 2024*, *arXiv:2401.12926* (1.3B matches 2x compute of random; similarity-to-quality can hurt). · Empirical data.

6. Implications for Data Strategy

First, value moves from the corpus to the curation function. If the marginal token can be negative, then a buyer's willingness to pay should track the quality of selection, not the gigabytes delivered. The asset that compounds is the

metric that ranks examples for inclusion or removal — exactly the artifact shown to convert power-law into near-exponential scaling [2] and, at LLM scale, to beat full-data training on a fraction of the tokens [7,8].

Second, the binding constraint is arriving on schedule. With effective human public text near 300T tokens and exhaustion projected for 2026-2032 [6], the industry cannot scale its way out; it must select its way out. Synthetic data, multimodal transfer, and data efficiency are the named escape hatches [6], and all three are selection problems — deciding which generated, transferred, or weighted examples earn a place in the budget. The synthetic-data hatch in particular is itself a selection problem: recursive training on model output collapses under a 'replace' workflow but is stabilized when real data is accumulated alongside synthetic, i.e. when the mixture is curated to retain real anchors [10]; and a systematic study across >1000 LLMs shows the payoff lives in the mixture ratio — roughly one-third rephrased-synthetic to two-thirds natural web text gives a 5-10x speedup to a target validation loss at larger budgets, while other ratios give back the gain, so 'use synthetic data' reduces to 'select the right synthetic/natural mixture' [15].

Third, selection must be conditioned on the deployment regime, not just intrinsic quality. The precision result means the 'right' corpus for an 8-bit served model differs from the right corpus for a 16-bit one [4]; the epoch result means the right corpus for a data-rich large model differs from that of a data-constrained one [3,5]; and DsDm shows the 'right' corpus is even conditioned on the target model itself, since model-aware selection beats model-agnostic similarity [8]. A curation service that exposes these conditioning levers sells a capability that static dataset vendors structurally cannot replicate.

Fourth, 'negative-value removal' is a demonstrable, billable outcome. A curator who can show that deleting an identified slice lowers deployed loss has a measurable product — the inverse of the volume vendor's pitch, and far harder to commoditize. The 2024 LLM-scale results make this concrete: 'we beat your full-corpus baseline using 30% of it' [7] and 'we matched 2x your compute on random data' [8] are demonstrations a buyer can independently reproduce on their own stack.

7. Investment Thesis

The scaling era trained the market to underwrite tonnage: crawl more, license more, label more. The evidence synthesized here says that strategy hits diminishing and then negative returns precisely as the supply of human text runs out [1,2,3,4,6,7,8]. That conjunction — flattening supply meeting a marginal-value curve that bends negative, now confirmed at LLM scale — is the setup for a category shift from data accumulation to data selection. The winners of the next cycle will be the firms that own the selection function.

The economics are asymmetric in the curator's favor. A volume vendor's marginal cost is roughly linear in tokens delivered and its product is commoditizing toward zero as the open web is exhausted and re-licensed. A selection vendor's product is a ranking-and-removal capability whose value grows with the size of the pool it filters [2] and with the buyer's deployment constraints [4], and whose best instances are cheap to run — a 125M reference model can curate data for a 3B target, an order-of-magnitude leverage on compute [9]. This is a capability with increasing returns and high switching costs, because it is validated against the buyer's own model and serving stack.

For Perpetual Labs the thesis is direct and defensible: build the curation/selection layer that can demonstrate negative-value removal — show a buyer that discarding an identified slice of their corpus lowers their deployed loss at their target precision, the way [7] and [8] do against full-data and random-selection baselines respectively. That is a claim a tonnage vendor literally cannot make, and it converts the macro data-exhaustion risk [6] from a headwind into the reason the product exists. The moat is the accumulated, regime-conditioned ranking metric — the one artifact in this literature shown to beat scaling itself [2] and to beat 2x compute at LLM scale [8].

We are explicit about what this thesis is and is not. It is a structural argument that a selection-capability business has better unit economics and defensibility than a volume-data business in a supply-exhausting market. It is not a claim about a specific buyer's budget or willingness-to-pay, which must be validated commercially. The market is every team that has already crawled the cheap web and is now staring at the negative tail of its own corpus — and, per DsDm, has discovered that naive quality-filtering does not necessarily beat random [8].



Figure 3. Accumulation vs. selection: strategic positioning of data products. Constructed 2x2 positioning four archetypes - raw web crawl (low/low), licensed bulk corpora, generic dedup/filter pipelines, and regime-conditioned selection (high/high). As supply exhausts and the marginal token bends negative, value migrates up-and-right toward selection functions that can demonstrate negative-value removal. Coordinates are illustrative, not measured. *Constructed strategic analysis by the authors; informed by [2,4,6,7,8]. Not empirical data. · Constructed analysis.*

8. Limitations and Counterarguments

Vision-to-language transfer is now partly, not fully, closed. The original concern — that the strongest pruning result [2] was demonstrated on vision benchmarks — is substantially answered by the 2024 LLM-scale selection results [7,8,9], which reproduce 'selection beats accumulation' at 1-3B parameters on web/Pile/C4/Dolma pretraining. The original LLM-scale results [7,8,9] are at the 1-3B scale; subsequent 2025-2026 work pushes data selection to genuine frontier scale — BETR's selection scaling laws extend to 10^{22} FLOPs [13], and a curated 20T-token corpus fed pretraining of a 400B-parameter frontier model [16] — so the 'sub-frontier only' caveat is now substantially relaxed for selection in general. What remains unproven at the frontier is the specific near-exponential scaling form [2]: that exact functional claim is still a vision result extrapolated, not demonstrated, at frontier LLM scale, and we flag it as such.

Regime dependence cuts both ways. The negative-marginal-token claim is conditional: in the unique-data, single-epoch, full-precision regime the marginal token remains positive and standard scaling laws hold. The thesis is that production increasingly leaves that regime, not that scaling laws are wrong within it.

Quantification gaps. The exact crossover D/N at which data becomes harmful [4], and the precise epoch at which decay turns negative rather than merely flat [5], depend on architecture, optimizer, and metric choices not fully pinned down in the source papers; our figures use reported anchor points (~4 and ~16 epochs; 7-8 compute-optimal

integer bits; 30% and 50% keep rates) rather than a closed-form curve. The supply projection [6] carries a wide stated confidence interval (100T-1000T tokens) and should be read as an order-of-magnitude constraint, not a date certain.

Metric dependence is the core execution risk. The selection advantage is entirely contingent on access to a high-quality ranking metric; most existing high-performing pruning metrics scale poorly [2], and DsDm shows that the intuitive similarity-to-quality metric may not beat random and can hurt [8]. The investment thesis therefore rests on the assumption that strong, regime-conditioned selection metrics can be built and improved — which is itself the product hypothesis being underwritten. A counterargument from the bull case for synthetic data also bears noting: if model-generated data can be cheaply manufactured and stabilized by accumulation [10], the 'supply exhaustion' pressure that makes selection binding could be partially relieved — though, as noted, choosing the synthetic mixture is itself a selection problem, so the demand for selection capability survives the objection.

9. Conclusion

The reflex of the scaling era — that more data is monotonically good — is a boundary-case truth that fails in exactly the regimes production cares about: repeated data, overtrained models, and quantized deployment. Across repetition [1], multi-epoch training [3,5], precision [4], classical pruning [2], and now LLM-scale selection [7,8,9], the marginal token's value is non-monotonic and can be zero or negative. The corollary is that selection, not accumulation, is the lever that still moves loss once the cheap web is exhausted [6] — and the 2024 evidence shows that lever working at language-model scale, where removing 70% of the data can help [7] and modeling the training process is worth 2x compute [8].

For a data-strategy organization the implication is clean: stop selling tonnage and start selling judgment. The defensible product is a regime-conditioned selection function that can demonstrate negative-value removal — provably making a model better by giving it less. That is the inversion this literature collectively forces, and it is the thesis on which the next data company should be built.

INVESTMENT THESIS

The scaling era taught the market to underwrite tonnage; this literature shows tonnage hits negative returns exactly as the human-text supply runs out — and, critically, the proof now holds at LLM scale, not just in vision. Five independent results break 'more is better': repetition double descent (0.1% of data repeated 100x degrades an 800M model to a 400M-equivalent) [1], multi-epoch decay (value of added compute decays to zero, ~16-epoch floor) [3,5], a quantization crossover where extra pretraining data becomes 'actively harmful' [4], pruning that approaches exponential scaling [2], and — the gap-closer — 2024 LLM-scale selection where pruning to 30% of the data beats the full corpus [7] and model-aware selection (DsDm) matches 2x the compute of random while naive quality-similarity filtering may not beat random and can hurt [8]. With effective public text near 300T tokens and exhaustion projected for 2026-2032 [6], you cannot scale your way out; you must select your way out. The defensible, increasing-returns asset is the curation/selection function — a regime-conditioned ranking metric that can demonstrate negative-value removal: prove to a buyer that deleting part of their corpus lowers their deployed loss at their target precision, or that you match 2x their compute on a curated subset. A tonnage vendor structurally cannot make that claim. For Perpetual Labs, data exhaustion is not a headwind — it is the reason the product exists. Underwrite the selection layer, not another crawler. The cheapest leverage is real: a 125M reference model can curate data for a 3B target [9]. And this is not a 2024 artifact: through 2025-2026 the selection lever scaled up, not down — benchmark-targeted ranking buys a 2.1x compute multiplier across 500+ models out to 10^{22} FLOPs [13], a regularized data-constrained recipe hits the loss asymptote with 5.17x less unique data [14], and a curated 20-trillion-token corpus set a new performance-vs-compute frontier and fed an actual 400B-parameter frontier model [16]. Selection is what reaches the frontier now.

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